

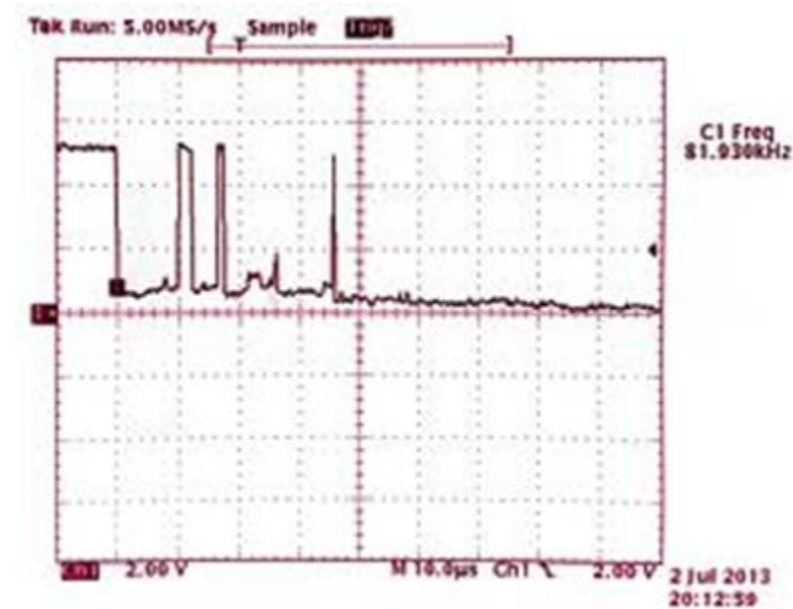
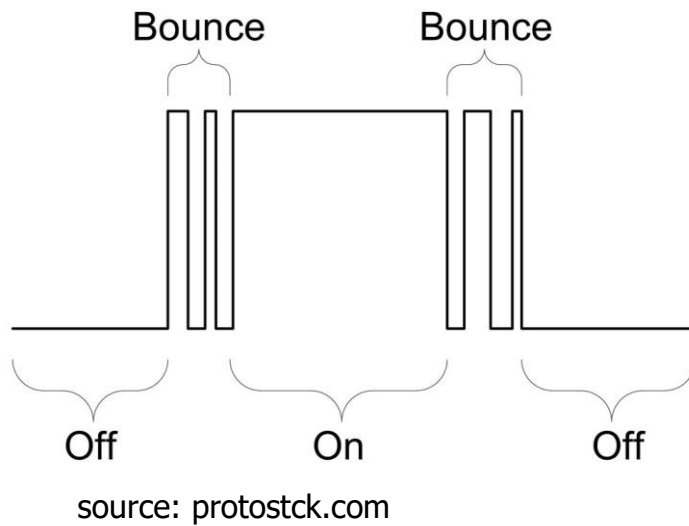


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FPGA Design Using VHDL

Module 5D: Debouncing

Button Debouncing 1



Button Debouncing 2

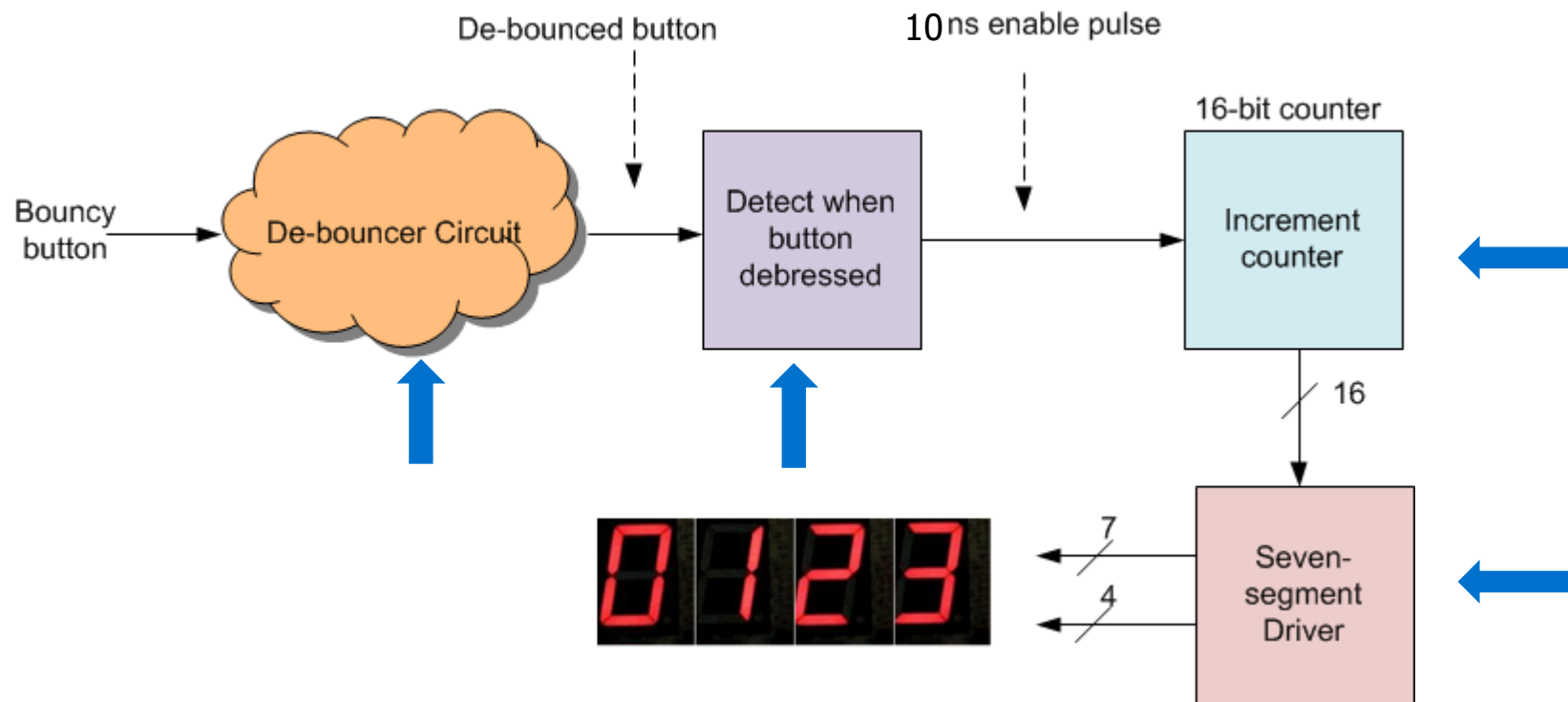
- Cannot simulate it easily, but it does exist
- Want to design a hardware component that eliminates bouncing
- Assume that you press the button for 0.1 seconds or longer
- One method to eliminate bouncing
 - Counter that increments as long as you press button
 - If counter $> 10,000,000$, then acknowledge press

Button Debouncing 3

```
if (pb0 = '1')  
    if (counter < 5000000)  
        increment counter  
    else  
        pb0db <= '1'  
else  
    pb0db <= '0'  
    counter <= 0's
```

How to Test Button Debounce

- Implement a design that tests your de-bouncer circuit
- Test to see if you de-bounced the pushbutton correctly



How to Test Button Debounce Continued

- Design the processes to generate a 20 ns pulse every time the push button is pressed down
- Design a process for a 16-bit counter with an enable that counts every time the enable goes high
- Display the 16-bit counter using the seven-segment display which you already designed from Lab 3